

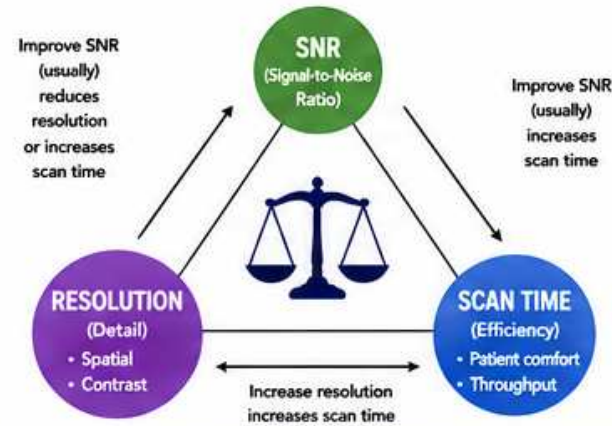
2.8 THE MRI IMAGE-QUALITY TRIANGLE

SNR ↔ RESOLUTION ↔ SCAN TIME

In MRI, you can improve two of the three, but the third will be affected.
The technologist's job is to find the right balance for the clinical question.

1 THE TRIANGLE

The Goal: Find the Best Balance



There is no perfect setting — only the best setting for the clinical task.

2 HOW MAJOR PARAMETERS PUSH THE TRIANGLE

Parameter	Effect on...			What It Does
	SNR	Resolution (Detail)	Scan Time	
NEX / Averages	↑↑	—	↑↑	More signal by repeating
Bandwidth (BW)	↓	—/↓*	↓	Higher BW = faster, but lower SNR
Matrix	↓	↑↑	↑↑	Higher matrix = more detail, longer time
FOV	↑	↓	—	Smaller FOV = more detail, lower SNR
Slice Thickness	↑	↓	—/↓*	Thinner slice = more detail, lower SNR, may be faster (less volume)
ETL / TSE Factor	↓	—	↓	Higher ETL = faster, but lower SNR
Parallel Imaging (R)	↓	—/↑*	↓	Faster scan, SNR penalty (more at higher R)
Field Strength (B ₀)	↑↑	—	—	Higher field = more SNR (also more artifacts)
Coil Elements	↑	—	—	More channels = more SNR (especially with PI)

* Can depend on other factors and how parameter is used.

3 WHAT HAPPENS WHEN YOU CHANGE ONE PARAMETER?

Goal	Typical Strategy	Trade-Off
Need better Image Quality (higher SNR)	• Increase NEX • Lower Bandwidth • Use larger coil / more channels • Use higher Field Strength	• Longer scan time • Potential artifacts (e.g., low BW)
Need more Detail (higher resolution)	• Increase Matrix • Decrease FOV • Decrease Slice Thickness	• Lower SNR • Longer scan time
Need Faster Scan	• Increase Bandwidth • Increase ETL • Use Parallel Imaging (higher R) • Decrease NEX	• Lower SNR • More artifacts • Lower resolution (if matrix reduced)
Need Balance (all-around)	• Optimize NEX, Matrix, BW, ETL • Use moderate Parallel Imaging • Match to clinical task	• Some compromise is unavoidable

Always match the balance to the clinical question — not just “better images”.

4 VISUAL EXAMPLES: MOVING ON THE TRIANGLE

A. Higher SNR Priority (But Longer Scan)

- NEX ↑
- BW ↓
- Matrix ↔
- ETL moderate
- Scan Time ↑↑

Best for: Subtle pathology (e.g., MS plaques, tumors)

B. Higher Resolution Priority (But Lower SNR & Longer Scan)

- Matrix ↑
- FOV ↓
- Slice thickness ↓
- NEX moderate
- Scan Time ↑↑

Best for: Small structures (e.g., cartilage, small lesions)

C. Faster Scan Priority (But Lower SNR)

- BW ↑
- ETL ↑↑
- Parallel Imaging ↑
- NEX ↓
- Scan Time ↓↓

Best for: Patient comfort, movement-prone exams



Both the protocol and the patient affect where you end up on the triangle.
Know the goal, understand the trade-offs, and make every parameter count.

5 EXAMPLE: SAME PATIENT, DIFFERENT BALANCE (KNEE PD FSE)

	High Quality (SNR Priority)	Balanced (General Use)	Fast Scan (Speed Priority)
NEX (Avg)	2	1	1
Matrix	512 x 512	320 x 320	256 x 256
BW (Hz/pixel)	31.2	62.5	125
ETL (TSE)	16	12	24
Parallel Imaging (R _s)	1 (Off)	2 (ARC)	3 (ARC)
Scan Time	6:20 min	3:10 min	1:20 min
SNR	★★★★★	★★★☆☆	★☆☆☆☆
Detail / Resolution	★★★★★	★★★☆☆	★☆☆☆☆

All images are same patient and sequence type.
Your choice depends on the clinical question and what you are trying to see.

6 RULES OF THUMB

- More signal = better images**
Increase NEX, lower BW, use larger coil, use higher field.
- More detail = higher matrix, smaller FOV, thinner slices**
Expect lower SNR and longer scan.
- Faster scan = higher BW, higher ETL, Parallel Imaging, fewer averages**
Expect lower SNR and more artifacts.
- Optimize for the question**
Not every exam needs “max everything”.
Balance is patient-specific and task-specific.



KEY TAKEAWAY

Master the triangle. Know the levers.
Make smart trade-offs. Deliver the right image the first time.

7 WHEN TO PRIORITIZE WHAT?

Clinical Situation	Priority
Detect small lesion / subtle disease	SNR / Resolution
Pre-surgical planning	Resolution
Motion-sensitive patient	Scan Time
Follow-up comparison	Consistency
High throughput / busy schedule	Scan Time
Research / quantitative imaging	SNR / Consistency

8 ENGINEER CHECKLIST

- ☒ Do I understand the clinical goal?
- ☒ Are the parameters balanced for that goal?
- ☒ Is SNR acceptable for the anatomy and coil?
- ☒ Is scan time reasonable for the patient?
- ☒ Are artifacts within acceptable limits?
- ☒ Can we improve quality without unnecessary time?



BALANCE IS THE ART. SCIENCE IS THE GUIDE. Understand the triangle. Use the tools. Trust your judgment.